

Report
Of
CORAL REEF BLEACHING PREDICTION DASHBOARD

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003003298

Name of Student: Shahnawaz Sheikh

Batch: 2025-29

CANDIDATE'S DECLARATION

I, Shahnawaz Sheikh, do hereby declare that the internship report titled "Coral Reef Bleaching Prediction Dashboard" has been prepared by me to fulfil the academic requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report describes the work and learning associated with the internship project. All references used in preparing this report have been acknowledged. I understand that I am responsible for the originality and accuracy of the work presented in this report.

Signature: _____

Student Name: Shahnawaz Sheikh

Roll No.: 25SCS1003003298

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the internship conducted under MARS - Center for Sustainable Community Development and the development of the Coral Reef Bleaching Prediction Dashboard. The internship provided an opportunity to understand how a real-world environmental problem can be studied using computer science, data analysis, machine learning, web technologies, and systematic software testing.

I am thankful to the mentors, faculty members, and team members who provided guidance, technical discussion, feedback, and support during the project. I am also grateful to the School of Computer Science and Engineering, IILM University, for providing the academic environment and opportunity to participate in this internship.

I would also like to acknowledge the public scientific and technical resources used by the project, especially coral-reef and oceanographic data services, and the open-source software ecosystem that supported the implementation of the dashboard.

Signature: _____

Student Name: Shahnawaz Sheikh

Roll No.: 25SCS1003003298

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/058

Date of Issue: July 31, 2026

This is to certify that

Shahnawaz Sheikh

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

Dr. Manish Prateek

Director
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INTERNSHIP OFFER LETTER



MARS

Center For Sustainable Community Development

Date: 30/05/2026

Subject: Offer Letter – MARS Centre for Sustainable Community Development Internship (01 June 2026 to 15 July 2026)

Dear Shahnavaz Sheikh,

We are pleased to welcome you to the MARS – Centre for Sustainable Community Development Internship Programme scheduled from 1st June 2026 to 15th July 2026. Your enrollment reflects your interest in creating meaningful impact through technology-driven sustainable development initiatives, and we are delighted to have you join this learning journey.

MARS – Centre for Sustainable Community Development is a Section 8 Company (Non-Government Organization) focused on creating technological solutions for sustainable development. One of the core missions of MARS is wildlife conservation, leveraging technology to protect and preserve biodiversity. The organization employs a range of innovative technological strategies including web development, mobile application development, extended reality (XR), new tools and techniques, and IoT devices to address real-world sustainability challenges.

At the beginning of the internship, you will receive training to help you understand the problem domains, field requirements, and the process of developing technology-based interventions. During the internship, you are required to maintain a daily diary of your work and observations, supported by geo-tagged photographs documenting your activities and field engagement.

The key problem statements for this internship include: (i) Life Below Water – developing solutions that support aquatic ecosystem conservation and monitoring; (ii) Life on Land – protecting biodiversity, forests, wildlife, and natural habitats; (iii) Sustainable Agriculture – improving agricultural productivity while conserving resources; and (iv) Eradication of Poverty through Technology – creating innovative technological interventions that improve livelihoods, accessibility, and socio-economic opportunities.

You will be expected to create computational and technology-enabled solutions for the challenges explained to you by domain experts. All projects will be undertaken under the guidance and mentorship of experts, enabling you to translate ideas into practical and scalable solutions.

The world urgently needs young innovators who can combine technology with sustainability to solve complex environmental and social challenges. Your contribution can help build resilient communities and create a more sustainable future for generations to come.

We look forward to your participation and wish you a productive and rewarding internship experience.

Yours sincerely,


Mrs. Priti Sinha (Director)
MARS- Center for Sustainable
Community Development

 marsgroup.cc@gmail.com

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4. PROJECT DESCRIPTION

4.1 Introduction

Coral reefs are among the most diverse marine ecosystems and provide habitat, coastal protection, food resources, and economic value to communities around the world. Coral bleaching occurs when environmental stress, especially unusually warm sea-surface temperatures, causes corals to lose the symbiotic algae that provide much of their energy and colour. If stressful conditions continue for a long period, bleaching can lead to disease and coral mortality.

The Coral Reef Bleaching Prediction Dashboard was developed as a browser-based decision-support platform that combines historical coral-reef observations, environmental parameters, machine-learning prediction, interactive visualisation, and climate-scenario analysis. The project uses a historical NOAA coral reef dataset spanning 1980-2020. After cleaning, 34,454 usable records are used for analysis, while twelve numerical environmental features are used by a Random Forest classifier to predict bleaching risk as Low, Medium, or High.

The dashboard presents summary statistics, five historical charts, an interactive world map of reef survey sites, a twelve-parameter prediction panel, prediction confidence and class probabilities, a plain-language assessment, recommended conservation action, and a ten-year climate-scenario simulator. It also includes a scoped reef assistant and a feedback mechanism.

My Role in the Project

My individual contribution focused mainly on research and testing. I researched existing coral reef monitoring dashboards and decision-support tools, analysed their features and limitations, reviewed the features included in our dashboard, and performed functional testing of the completed system. I did not independently develop the complete machine-learning model or the complete full-stack application.

4.2 Organization Profile

The project was completed as part of the internship conducted under MARS - Center for Sustainable Community Development. The internship provided a project-oriented learning environment in which students worked in teams on sustainability-related real-world problem statements and applied computational thinking and software-development concepts.

During the internship, the team worked in the domain of marine ecology and coral-reef conservation. The project required the team to understand an environmental problem, study existing solutions, identify suitable technical approaches, develop a dashboard-based solution, and test the completed system.

MARS - Center for Sustainable Community Development conducted the internship program, which focused on sustainability-oriented problem solving. The exact internship duration, mentor name, internship dates, and certificate details should be inserted from the official MARS internship certificate or communication before final submission.

- Internship Program: MARS - Center for Sustainable Community Development
- Organization / Center: MARS - Center for Sustainable Community Development
- Internship Domain: Computer Science applied to marine ecology and coral-reef conservation
- Project: Coral Reef Bleaching Prediction Dashboard
- Internship Duration: 1june2026 - 15july2026
- Mode / Location: Virtual

4.3 Problem Statement

Most coral-reef monitoring systems are strong at describing current or recent thermal stress, but conservation teams often need an easier way to interpret environmental measurements as an actionable bleaching-risk level. Historical bleaching records, satellite observations, geographic reef information, and long-term climate projections can also be spread across different tools, making them difficult for non-specialist users to interpret together.

The project addresses this gap by creating a single dashboard that brings together historical analysis, reef-location visualisation, live oceanographic information, machine-learning-based risk classification, plain-language assessment, recommended action, and scenario-based projection.

- Reactive monitoring can identify stress after it has already accumulated, reducing the time available for preventive action.
- Historical data, satellite data, and geographic reef information are often fragmented across different sources.
- Raw variables such as SST, SSTA, and DHW can be difficult for non-specialist users to interpret directly.
- Users need an understandable Low / Medium / High risk classification with confidence and supporting explanation.
- Long-term conservation planning benefits from a simple way to compare alternative climate trajectories.

4.4Project Objectives

The main objective of the project is to create an accessible early-warning and decision-support dashboard for coral bleaching risk. The system is designed to convert environmental data and historical observations into understandable information that can support researchers, conservation teams, students, and other interested users.

- Analyse a large historical coral-reef dataset and present important bleaching patterns through interactive charts.
- Use twelve environmental predictor features for machine-learning-based bleaching-risk classification.
- Predict coral bleaching risk as Low, Medium, or High and display model confidence and class probabilities.
- Provide a visual world map of historical reef survey locations and bleaching severity.
- Use live NOAA oceanographic data for monitored reef zones where available.
- Provide a ten-year scenario simulator for Optimistic, Status Quo, and Worst-Case climate trajectories.
- Explain prediction results in plain language and provide a recommended action.
- Provide an interface that is understandable to users without advanced programming or data-science knowledge.
- Evaluate the completed dashboard through functional testing.

4.5 Scope of the Project

The scope of the Coral Reef Bleaching Prediction Dashboard covers historical analysis, environmental-parameter-based prediction, interactive visualisation, live ocean-data retrieval for selected monitored zones, climate-scenario simulation, AI-assisted explanation, and user feedback.

- Historical NOAA coral-reef records from 1980-2020 are used for analysis and model development.
- The cleaned working dataset contains 34,454 records.
- Twelve environmental parameters are used for bleaching-risk prediction.
- The dashboard displays historical summary cards and five interactive charts.
- An interactive map displays more than 8,600 historical reef survey sites.
- The prediction module returns Low / Medium / High risk, confidence, class probabilities, assessment, and recommended action.
- The scenario simulator projects Reef Health % and Bleaching Probability % for ten years under three climate scenarios.
- The project is a web-based academic decision-support system; it does not directly deploy underwater IoT sensors or perform direct operational intervention at reef sites.

Individual Contribution Scope

My contribution was limited to research on existing coral reef dashboards, feature comparison and analysis, review of the completed dashboard, and functional testing. Team-level development activities described elsewhere in this report represent the complete project and should not be interpreted as work completed individually by me.

4.6 Technologies and Tools Used

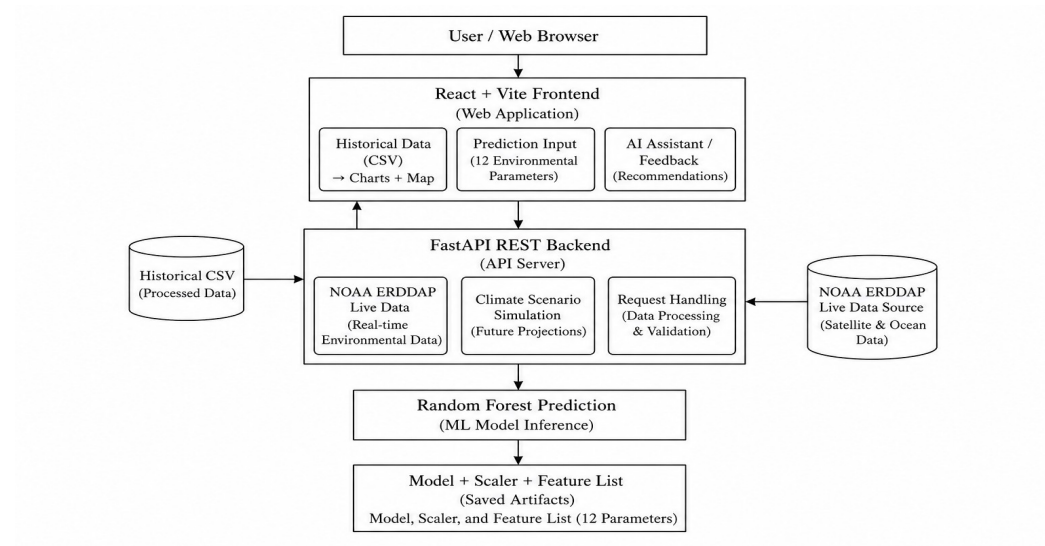
Layer / Purpose	Technology / Tool
Frontend	React 18, Vite, Tailwind CSS
Charts	Recharts
Interactive Map	Leaflet / React-Leaflet
Backend	Python, FastAPI, Uvicorn
Machine Learning	scikit-learn Random Forest Classifier
Class Balancing	imbalanced-learn (SMOTE)
Data Processing	pandas, NumPy
Live Ocean Data	NOAA ERDDAP
Feedback Storage	Supabase
AI Explanation	LLM API provider
Geolocation	Browser/IP-based geolocation API
Deployment	Vercel frontend and Render backend

Python and scikit-learn support the machine-learning pipeline. The backend uses FastAPI to expose REST endpoints for health checking, live data, prediction, and simulation. React is used to build the single-page frontend. Recharts and Leaflet provide data visualisation and mapping. Supabase stores feedback submissions, while NOAA ERDDAP provides live oceanographic data for selected reef zones.

4.7 System Architecture

The application follows a client-server architecture. The React frontend handles presentation and user interaction, while the FastAPI backend handles machine-learning inference, climate simulation, and live NOAA data retrieval. Historical data and trained model artifacts form the data and machine-learning layers.

SYSTEM ARCHITECTURE



The backend exposes GET /health for server and model health, GET /live-data for live NOAA information, POST /predict for bleaching-risk prediction, and POST /simulate for multi-year scenario simulation. The frontend sends validated user inputs to these endpoints and renders the returned information as cards, probability bars, assessments, maps, and charts.

4.8 Methodology

The project follows a structured pipeline from data collection to deployment and testing.

Step 1 - Data Collection

The project uses a NOAA coral reef dataset containing approximately 41,361 raw records collected from 1980 to 2020.

Step 2 - Data Cleaning and Preprocessing

Duplicate records are removed, selected columns are converted to numeric values, rows without a usable target are removed, and missing feature values are filled using column medians. The resulting cleaned dataset contains 34,454 records.

Step 3 - Feature Preparation

Twelve environmental features are selected: Temperature Mean Celsius, SSTA, SSTA DHW, TSA DHW, Turbidity, Cyclone Frequency, Windspeed, Depth, SSTA Frequency Max, TSA Mean, Climatic SST, and TSA Frequency.

Step 4 - Risk Label Creation

Percent Bleaching is converted into Low, Medium, and High risk classes for classification.

Step 5 - Scaling and Class Balancing

StandardScaler is used on the numeric features and SMOTE is applied to the training data to reduce class imbalance.

Step 6 - Model Training

A Random Forest classifier is trained using 200 trees with a maximum depth of 15 and a minimum of 5 samples required for an internal split.

Step 7 - Evaluation and Model Saving

The model is evaluated on a held-out test set. The trained model, scaler, and ordered feature list are saved for use by the live backend.

Step 8 - Web Application Development

The React frontend communicates with the FastAPI backend to obtain predictions, simulations, and live data and to display historical charts and the interactive map.

Step 9 - Functional Testing

The completed dashboard is tested by interacting with user-visible functions and comparing the actual result with the expected result.

Functional Testing Performed

ID	Function	Action	Expected Result	Actual / Status
FT-01	Dashboard loading	Open the application	Loading screen redirects to the main dashboard	[TO BE RECORDED]
FT-02	Summary statistics	Verify the four summary cards	All historical summary values are displayed	[TO BE RECORDED]
FT-03	Historical charts	Inspect the five historical charts	Charts render with historical data	[TO BE RECORDED]
FT-04	Interactive map	Zoom, pan and inspect survey points	Map responds and survey points remain visible	[TO BE RECORDED]
FT-05	Prediction inputs	Adjust the twelve environmental sliders	Each input value changes correctly	[TO BE RECORDED]
FT-06	Bleaching prediction	Click Predict Bleaching Risk with valid inputs	Risk class, confidence and probabilities are returned	[TO BE RECORDED]
FT-07	Assessment and action	Review prediction output	Assessment and recommended action are displayed	[TO BE RECORDED]
FT-08	Climate scenario simulator	Select Optimistic, Status Quo and Worst Case	Ten-year projection changes for the selected scenario	[TO BE RECORDED]
FT-09	Reef Assistant	Ask a coral-reef-related question	A relevant plain-language response is returned	[TO BE RECORDED]
FT-10	Feedback form	Enter valid feedback and submit	Submission is accepted by the feedback module	[TO BE RECORDED]

The Actual / Status column should be completed using the results observed during the final testing session. Screenshots may be attached as evidence. Test results should not be marked Pass unless the corresponding function has actually been executed and verified.

4.9 Expected Outcomes

- A functional web dashboard that presents historical coral bleaching information in an accessible format.
- A cleaned historical dataset suitable for analysis and machine-learning prediction.
- A Random Forest model that classifies bleaching risk into Low, Medium, and High categories from twelve environmental inputs.
- Interactive historical charts and a global reef survey map for visual exploration.
- Prediction output containing risk class, confidence, class probabilities, assessment, and recommended action.
- A ten-year scenario simulator that compares Optimistic, Status Quo, and Worst-Case trajectories.
- A scoped AI assistant that helps users understand reef-related prediction results.
- A feedback mechanism for collecting user ratings and comments.
- Improved understanding of how computer science and machine learning can be applied to environmental conservation problems.

Learning Outcome from My Contribution

Through my research and testing role, I developed a better understanding of existing coral reef monitoring tools, feature comparison, software requirement analysis, dashboard evaluation, and systematic functional testing. The work also improved my ability to understand a technical project beyond only its source code by connecting the problem statement, user requirements, system features, and observed software behaviour.

4.10 Certificates of Completion and Other Communication Proof



MARS - Center for Sustainable Community Development <marsgroup.cc@gmail.com>

Aug 2, 2026, 8:29 AM ☆ ☺ ↶ ⋮

to Krishnavijay2111, prashant.25scs1003005064, ananya5005sharma, dsirari7689, Krishna.25scs1003001057, ekansh.21593, mishra.25scs1003001160, roshan.25scs1003001159, me ▾

Dear Intern(s),

Congratulations on successfully completing your internship with **MARS – Centre for Sustainable Community Development!**

We sincerely appreciate your dedication, enthusiasm, and the valuable contributions you made during your internship. Your willingness to apply your computational skills to address real world social challenges reflects the spirit of innovation and service that MARS strives to promote.

As you move forward in your academic and professional journey, we encourage you to continue using your knowledge and technical expertise to create meaningful solutions for society. Technology has the power to transform lives when guided by compassion, responsibility, and a commitment to sustainable development. We hope this internship has inspired you to become not only skilled professionals but also socially conscious changemakers.

Please find your **Internship Completion Certificate** attached to this email. We congratulate you once again on this achievement and wish you continued success in all your future endeavors.

Thank you for being a part of the MARS family. We look forward to seeing the positive impact you will create in the years ahead.

With best wishes,

Team MARS

MARS – Centre for Sustainable Community Development

5. BIBLIOGRAPHY / REFERENCES

1. National Oceanic and Atmospheric Administration (NOAA), Coral Reef Watch resources and coral reef environmental data used by the project.
2. Coral Reef Bleaching Prediction Dashboard - project report supplied by the project team, academic session 2025-2026.
3. scikit-learn documentation - RandomForestClassifier and StandardScaler.
4. imbalanced-learn documentation - Synthetic Minority Over-sampling Technique (SMOTE).
5. FastAPI documentation - Python web API framework.
6. React documentation - frontend user-interface library.
7. Leaflet documentation - interactive web mapping library.
8. Recharts documentation - charting library for React.
9. Supabase documentation - backend service used for feedback persistence.